How does Convolutional Neural Network (CNN) compare to a RS statistical test in accurately and reliably detecting LSB steganography in PNG images?

Gullbrandsson, Martin

Pakos, Jan

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Supervisor: Johannes Olegård
Department of Computer
and Systems Sciences

Stockholm
University

Abstract

Steganalysis has typically relied on hand-crafted statistical means of discovering concealed data, however, in recent years the effort has largely shifted towards automated feature extraction techniques via deep learning. Reliable detection of such hidden content is of direct relevance to law enforcement and cybersecurity practitioners tasked with uncovering covert communication channels in digital forensic investigations. This thesis examines whether the newer Convolutional Neural Networks (CNNs) are more effective than traditional RS-analysis methods to detect Least Significant Bit (LSB) steganography utilized to hide data in lossless PNG image files. To evaluate these two detection methods, a quantitative experiment was performed using BOSSBase 1.01 dataset on 4 different embedding techniques: Sequential LSB, Randomized LSB, Edge-Adapted LSB, and LSB Matching, at three different payload capacities of 25%, 50%, and 75%. The results of this investigation show that traditional RS-analysis or modern CNN methods can be optimal based on what embedding strategy has been used. RS-analysis was superior to CNN (SRNet) for both Sequential and Randomized LSB modifications across all examined payloads, achieving greater than 95% accuracy for both modified versions even at the lowest embedding rate of 25%. Conversely, for low payload Edge-Adaptive LSB, CNN demonstrated superior robustness with an accuracy of 79.12% versus the RS-analysis's accuracy of 72.50% at a 25% payload. Finally, only the CNN was capable of detecting LSB Matching (with an accuracy of 91.36%), where the RS-analysis performed below chance level with its ability to detect the embedding method. Therefore, these results demonstrate that traditional statistical techniques continue to be very effective and sufficient for the very basic Least Significant Bit substitution; however, deep learning techniques are needed for detecting Edge-Adaptive LSB and/or LSB Matching steganography.

Acknowledgements

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We are also sincerely thankful to the faculty and staff of the Department of Computer and Systems Sciences, whose support and resources made this research possible.

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DSR	Design Science Research
FPR	False Positive Rate
GNCNN	Gaussian-Neuron Convolutional Neural Network
LSB	Least Significant Bit
ML	Machine Learning
NLP	Natural Language Processing
PRNG	Pseudo-Random Number Generator
ROC	Receiver Operating Characteristic
RQ	Research Question
RS	Regular-Singular
SRNet	Steganalysis Residual Network
SQ	Sub-question
TPR	True Positive Rate
Bpp	Bits Per Pixel

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1

Introduction

In the age of widespread digital communication and increasing concerns about data privacy, the methods of concealing information have become vital in computer and systems sciences research. Steganography, derived from Greek words “covered” and “writing”, is defined as the practice of concealing information inside, often innocent, digital media, such as images, for secret communication [9]. While cryptography encrypts information in order to make it unreadable, steganography is concerned with concealing the very existence of the information [9], making it valuable for scenarios where the attention drawn to encrypted messages would be undesirable.

Steganographic methods have a number of serious implications concerning security and forensics. Although they can be used for legal purposes, such as digital watermarking used for copyright protection, secure communication in countries that ban or restrict encryption, and privacy preservation, these methods can be used for other malicious tasks as well [12], which makes reliable detection a matter of broader societal interest. Research into steganalysis therefore contributes directly to the work of forensic specialists and cybersecurity practitioners tasked with uncovering hidden communication channels that would otherwise evade conventional inspection.

The countermeasure to steganography is known as steganalysis. An “arms race” has developed many techniques against steganography. These techniques aim to develop reliable ways to detect the presence of hidden data in digital media. The detection methods fall primarily into two categories.

The more traditional ones include statistical analysis, such as the Regular-Singular (RS) analysis, which detects the disruptions in spatial correlations and local smoothness that random LSB embedding introduces to the pixel data [17].

The more recent, second category uses machine learning, more specifically deep learning, where models such as Convolutional Neural Networks (CNNs) are trained to detect the fine, high-dimensional patterns and noise artifacts that are present in stego images [19].

The advancements in deep learning research have resulted in a paradigm shift for steganalysis research with a tendency to regard Neural Networks as the norm for solving steganalytic detection tasks. However, this paradigm shift may tend to neglect the computational complexities and detection accuracies. Although the use of Convolutional Neural Networks (CNNs) with state-of-the-art performances for complex and adaptive steganographic attacks is compelling, their superiority over traditional methods for basic LSB embedding schemes within lossless images is still worthy of research.

This study compares the two approaches, and aims to provide a direct and empirical comparison of their effectiveness.

1.1 Delimitations

1.1.1 Image file format (PNG Only)

The study will focus on Least Significant Bit (LSB) steganography, which requires exact pixel value preservation. Lossy image formats, such as JPEG, destroy LSB information in the compression process and introduce artifacts that interfere with the specific statistical analysis methods being tested.

1.1.2 Spatial Domain LSB Substitution

This research aims to compare the baseline effectiveness of the statistical method (RS-analysis) against the Deep Learning model (CNN). The statistical test is designed to detect spatial anomalies in LSBs. Including other adaptive methods, would require different steganalytic methods, due to how they work, which is outside of the scope of this comparison.

1.1.3 Dataset and Color Channels

The BOSSBase dataset is widely used, ensuring reproducibility [20]. The images are in grayscale, the embedding artifacts are then isolated in luminance values.

1.1.4 Payload Granularity

This study opts for discrete payload sizes rather than a continuous spectrum. In order to answer the sub-question regarding changes as payload decreases, the study tests fixed embedding rates (25%, 50%, and 75% of image embedding capacity for respective steganographic method) to provide clear comparative bins for the True Positive Rate and detection accuracy analysis.

1.2 Research Question

This study is an empirical comparison, intent on investigating the effectiveness of traditional statistical steganalysis versus modern deep learning classification in detecting steganographic embedding.

This is condensed into the following **Research Question (RQ)**:

> **RQ:** How does Convolutional Neural Network (CNN) compare to a RS-Analysis statistical test in accurately and reliably detecting LSB steganography in PNG images?

With the help of the following **Sub-Question (SQ)**:

> **SQ :** How does the detection accuracy of each method change as the size of the hidden payload decreases?

2

Extended Background

2.1 Literature Review

2.1.1 Fundamentals of LSB Steganography

LSB embedding stands out as one of the most popular and accessible steganographic techniques. LSB steganography replaces the value of a least significant bit of a pixel with the bit from the hidden message, as described by Chan and Cheng [3]. For instance, in an 8-bit grayscale image, modifying the least significant bit of a pixel value from 10110110 (182) to 10110111 (183) results in only one change in intensity level (see figure 2.1), which is essentially undetectable by the human eye. The advantage of LSB insertion lies in its simplicity and high embedding capacity; however, these advantages also exhibit statistical abnormalities that can be utilized in the detection process. The image format used to contain the hidden information plays a vital role in both the effectiveness of the steganographic embedding and its detectability. While not the only lossless format, PNG (Portable Network Graphics) images are highly relevant to LSB steganography given their characteristics as a lossless compressible file format. While JPEG is a useful format, it is lossy and often destroys hidden data when compression artifacts are introduced [7]. PNG images and their deflate compression algorithm preserve pixel values exactly, thus maintaining an ideal medium for LSB steganography because the image maintains the exact bit values required for message recovery.

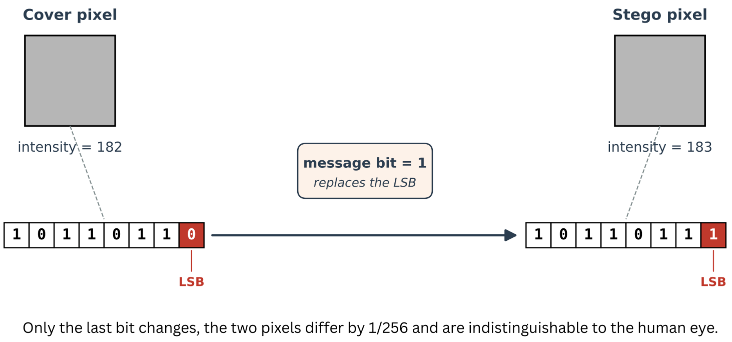

Figure 2.1: LSB Illustration.

2.1.2 Traditional Steganalysis

Fridrich, Goljan, and Du [10] introduced the RS (Regular-Singular) steganalysis method to detect LSB steganography in grayscale images. Unlike histogram attacks (like Chi-Square) that look for global frequency anomalies, RS analysis uses spatial correlations between groups of pixels to estimate the initial cover image.

RS analysis works by dividing the image into disjoint groups of n adjacent pixels (e.g., 2×2 blocks). A discrimination function, f, is defined to measure the "noisiness" or variation within each group. As defined by Fridrich, Goljan, and Du [10], for a group of pixels $G = (x_1, x_2, \ldots, x_n)$, the discrimination function is calculated as the sum of absolute differences between adjacent pixels:

$$f(x_1, x_2, \ldots, x_n) = \sum_{i=1}^{n-1} |x_{i+1} - x_i| \tag{2.1}$$

By applying a flipping operation (mask M) to the LSBs of the pixels in a group, the block's noisiness changes. Based on how the discrimination function value changes after flipping (denoted as $F_M(G)$), groups are categorized into:

- **Regular (R):** Flipping increases the noisiness ($f(F_M(G)) > f(G)$).
- **Singular (S):** Flipping decreases the noisiness ($f(F_M(G)) < f(G)$).
- **Unusable (U):** Flipping does not change the noisiness ($f(F_M(G)) = f(G)$).

Statistical symmetry is the core premise of RS analysis. In an image with no embedding, the number of Regular Groups (R_M) should be approximately equal to the number of Regular groups obtained using an inverse mask (R_{-M}); the case is similar for Singular groups:

$$R_M \approx R_{-M} \quad \text{and} \quad S_M \approx S_{-M} \tag{2.2}$$

The natural balance is disturbed by LSB embedding, and the disturbance grows as the payload increases. As the size of this modified payload increases, the difference between (R_M) and (S_M) will become smaller. This process continues until the image becomes completely randomized; at that point, R_M and S_M converge. Since RS Analysis measures absolute deviation from this perfect symmetry, it not only allows for detection of steganography but also allows for a more accurate estimation of embedded message length than earlier first-order statistical tests.

2.1.3 Deep Learning in Steganalysis

The limitations of handcrafted statistical features, such as the rigid assumptions regarding spatial correlation in RS-analysis, paved the way for deep learning approaches. While RS-analysis relies on explicit knowledge of the embedding process, deep learning (DL) methods can automatically learn complex patterns that distinguish cover images from stego-images. This capability gives DL methods an advantage, as a Convolutional Neural Network (CNN) can learn to identify steganographic artifacts from raw data that remain statistically invisible to fixed-model testing methods [8].

The field has experienced a paradigm shift from spatial-domain analysis toward adaptive domain techniques driven by artificial intelligence [1]. The primary method that has emerged is the use of CNNs. However, the implementation of CNNs for steganalysis provides unique challenges compared to traditional computer vision tasks. In a classic example of image recognition (e.g., recognizing a cat in a photo), the object being identified is the dominant signal.

Steganographic modifications, conversely, are designed to be imperceptible, hidden behind the “strong” noise of the image content [19]. Standard

CNN techniques usually fail to detect these weak signals because the high-amplitude content suppresses them. To address this, Qian et al. [19] proposed a customized deep learning model known as the Gaussian-Neuron CNN (GNCNN).

The GNCNN introduces a pre-processing layer designed to suppress image content and enhance the signal-to-noise ratio of the steganographic embedding. A high-pass filter is applied before the data enters the convolutional layers, effectively removing the "cover image" and leaving the "noise residual" where steganographic artifacts reside. This is similar to the residual calculation step in traditional steganalysis but is integrated directly into the learning pipeline.

The key benefit of using CNNs over traditional statistical methods, such as RS-analysis, is that the CNN architecture combines both feature extraction and classification into one unified structure. In traditional methods, feature extraction is completed first, and those features are used to train the classifier independently; with CNNs, both feature extraction and classification occur in concert using backpropagation [19].

2.1.4 Sensitivity and Robustness

Researching payload size raises issues regarding detection sensitivity. RS-Analysis relies on comparing variations within block statistics (based on R_M versus S_M). As the payload decreases, the difference becomes negligible. At low payload size (below 0.05 bpp), the modification to pixel groups does not create enough signal to differentiate the image from natural noise. While statistical approaches struggle to create an accurate threshold in these low-payload situations, deep learning models have shown superior performance [1].

Combating the false positive rate is also vital for operational security. A key weakness of statistical tests is the occurrence of false alarms when viewing noisy images or those with distinct textures. Scanned photographs or images altered using specific filters often possess statistical properties that confuse RS steganalysis, incorrectly classifying clean images as stego-images [9]. CNNs typically offer better generalizability. After being trained on large datasets, deep learning models can learn to detect the specific artificial noise introduced by LSB embedding, reducing the false positive rate compared to rigid statistical tests [19] [1].

3

Method

To address the research question, this study employs a quantitative experimental setup. In line with the description and definition of an experiment by Denscombe [5], this study manipulated independent variables in order to measure their impact on the dependent variables. This was done for each steganalysis method (RS-Analysis and CNN), and the yielded observations were compared in order to draw conclusions concerning the relative effectiveness of the steganalysis methods.

In this study, the steganographic embedding method and the payload size were used as independent variables, and detection accuracy, true positive rate, and false positive rate as the dependent variables.

3.1 Ethical Considerations

This study relies entirely on the processing of pre-existing digital data and does not involve human participants. Thus, the ethical concerns regarding informed consent, anonymity, and voluntary participation are not applicable [5]. The BOSSBase 1.01 dataset [20] is a standard academic image library recognized in forensics research. No personal or sensitive data was collected or generated during the experimental process.

The key ethical consideration of this study relates to the "dual-use" of steganography and steganalysis. Steganography can be applied for ethical purposes like protecting privacy and copyrights, it can also be used for malicious purposes like data exfiltration attacks and command and control communication in cyber-attacks. Studying steganalysis, or techniques for detecting steganographic communications, is a positive contribution as it helps forensic specialists in detecting hidden communication channels. Instead of using the research for developing new embedding algorithms that are difficult to be detected, this study sticks to the detection of LSB steganography, which is an example of good security research ethics.

To ensure scientific integrity, this study takes a the importance of replicability into consideration. Using a public data set with standard evaluation criteria (True Positive Rate, True Negative Rate, False Positive Rate, False Negative Rate, Accuracy) helps ensure that the comparison between the CNN and RS-Analysis is sound, so that the capability of the deep learning approach is not misrepresented for security applications.

Further ethical implications of this study are discussed in section 5.

3.2 Experimental Design

The conduct of the experiment can be examined as a four-step process.

1. Cover Image Selection and Preparation.
2. Stego-Image Embedding.
3. Implementation of Steganalysis
4. Performance Evaluation.

3.2.1 Cover Image Preparation

Dataset

BOSSBase 1.01 was selected due to its established prevalence in the field of steganography research [20]. While the original dataset images were provided in Portable Graymap (.PGM) format, these images were subsequently converted into Portable Network Graphics (.PNG) format. This conversion maintained the lossless nature of the images while providing a closer resemblance to real-world digital environments, where PNG files are significantly more common [22].

3.2.2 Stego-Image Generation

Payload Sizes (Embedding Ratios)

For the size of the embedding message 25% ($r = 0.25$), 50% ($r = 0.50$), and 75% ($r = 0.75$) of the embedding capacity was chosen.

LSB Methods

Four different LSB-based steganographic methods of varying sophistication were selected for comparative testing. This was done with the goal of increasing the generalizability of the assessment. These were implemented in Python scripts from the theoretical sources listed in Table 3.1.

METHOD	DESCRIPTION	SOURCES
`Sequential LSB`	LSB embedding in its simplest form.	Chan and Cheng [3]
`Randomized LSB`	Use of PRNG for selecting embedding locations.	Sharp [21]
`LSB Matching (±1)`	Randomly adds or subtracts 1 from the pixel value instead of replacing it.	Ker [14]
`Edge Adaptive LSB`	Identifies “noisy” pixels for embedding.	Luo, Huang, and Huang [16]

Table 3.1: LSB Methods in this Study

3.2.3 Steganalysis Implementation

RS-Analysis

This steganalysis method was implemented in Python, adhering to the specifications outlined by Fridrich, Goljan, and Du [10].

Convolutional Neural Network (CNN)

For the ML model, SRNet (Steganalysis Residual Network) [2] was chosen. This was motivated by its architecture, which aims to extract noise residuals, where the hidden messages might reside, in early layers, before they become suppressed in successive layers [23], and its documented performance detecting stego images embedded with advanced steganographic methods [2].

SRNet Implementation and Training Details

To ensure full reproducibility, the specific hyperparameters and training configurations utilized for the SRNet model are detailed below.

Weight Initialization The convolutional layers of the network were initialized using a Xavier uniform distribution, with biases set to a constant value of 0.2. The fully connected (Linear) layers were initialized using a Normal distribution ($\mu = 0.0, \sigma = 0.01$) with biases set to 0.0.

Optimizer and Loss Function The network was trained using the Adamax optimizer. The optimizer hyperparameters were configured to $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with no weight decay applied. The final layer utilizes a LogSoftmax activation, and the model was optimized against a Negative Log-Likelihood Loss (NLLLoss) function.

Training Pipeline and Augmentation A batch size of 20 was utilized during training. The initial learning rate was set to 0.001 and was decayed by a factor of 0.1 every 30 epochs. To increase robustness, data augmentation was applied to the training set by randomly rotating the input images by $90°$.

Table 3.2: SRNet training configuration.

Parameter	**Value**
Input size	512×512 grayscale
Train / Val / Test	7000 / 1500 / 1500
Batch size	16 cover–stego pairs (32 images)
Optimizer	Adamax
Initial learning rate	1×10^{-3}
LR schedule	$\times 0.1$ on validation plateau
Loss	Binary cross-entropy
Max epochs	30
Early stopping	patience 10, min. delta 1×10^{-4}
Augmentation	90° rotations, horizontal/vertical flips
Random seed	42
Framework	PyTorch 2.x

3.2.4 Performance Metrics

The results of the experiment were compiled into two tables with several relevant metrics, grouped by embedding technique.

3.3 Alternative Research Strategies and Methods

3.3.1 Choice of Dataset

Sedighi, Fridrich, and Cogranne [20] concludes that the choice of cover images has a huge impact on the effectiveness of different steganographic and steganalysis methods. Evaluating effectiveness of these methods on the same cover image source might thus not yield a complete generalizable result. An alternative approach would be to have the cover image source as an independent variable, although this would enlarge the study considerably.

Another approach would be to focus the study on certain fixed scenarios where cover source and steganographic method are predefined to be the presumed optimal for each other, and then compare different steganalysis methods.

3.3.2 Other Steganographic Methods

This study does not include more modern steganographic methods like WOW, HUGO, MiPOD, or SUniward. This was done to limit the study, but decreases its usability in a modern context. Another approach could include or focus solely on one or more of these methods.

3.3.3 Design Science Research

One possible alternative approach that might have been adopted is Design Science Research (DSR), which has been formally introduced by Chatterjee and Hevner [4]. In DSR, the focus is on developing and assessing a new artifact, such as a new detection algorithm, an improved CNN framework, or an integrated detection scheme based on statistics and deep learning, with the designed artifact being the key research contribution. In this process, knowledge creation occurs by repeated building and testing, followed by communicating design knowledge as the research outcome. DSR was taken into consideration but eventually rejected as the main method for use in the current thesis. This is due to the nature of the research question asked, which is comparative in its approach rather than constructive.

3.3.4 The Use of AI tools

To meet the requirements of academic integrity, the authors wish to acknowledge that AI tools were used in the process of writing this thesis in a supporting role. In particular, AI programs were used for: verifying the spelling and grammar in sentences, proposing rewrites aimed at enhancing the readability of sentences written by the authors, helping find and pinpoint source literature, which was then reviewed, validated, and referenced by the authors directly, and giving support in coding the experiment pipeline, involving debugging and syntax.

4

Results

RS estimates were computed for both clean and stego images across three embedding rates ($r \in \{0.25, 0.50, 0.75\}$) in order to evaluate the detectability of each embedding technique. For clean images, the mean RS estimated embedding rate was 0.0211 (actual rate was zero) with a standard deviation of 0.0842.

A binary classifier was constructed by thresholding the RS estimate at a value τ. An image whose estimated embedding rate met or exceeded τ was classified as stego; otherwise it was classified as cover. This yields four outcomes: a stego image correctly classified as stego is a True Positive (TP), while one falling below τ is a False Negative (FN). A cover image correctly classified as cover is a True Negative (TN), while one meeting or exceeding τ is a False Positive (FP). The threshold τ was selected to be the embedding rate for each of the three: 0.25, 0.50 and 0.75.

4.0.1 ROC & AUC

In both steganalysis algorithms described above, we obtain a score per image reflecting its likelihood of carrying concealed data. To turn this score into a binary decision, we have to choose a threshold. An image with a score above the threshold is considered stego; otherwise, it is classified as clean. Setting the threshold is a balancing act: a lower threshold increases true positives (TP) at the expense of false positives (FP), while a higher

Table 4.1: Classification performance of RS analysis for different LSB embedding techniques and rates.

Strategy	Rate	AUC	Accuracy	Threshold τ	TP	TN	FP	FN
Sequential	0.25	0.982	95.4%	0.25	1441	1420	80	59
	0.50	0.998	98.6%	0.50	1482	1477	23	18
	0.75	0.999	99.8%	0.75	1497	1496	4	3
Random	0.25	0.986	97.0%	0.25	1469	1440	60	31
	0.50	0.999	99.5%	0.50	1497	1487	13	3
	0.75	0.999	99.9%	0.75	1499	1498	2	1
Adaptive	0.25	0.761	72.5%	0.25	1095	1081	419	405
	0.50	0.954	90.2%	0.50	1385	1321	179	115
	0.75	0.997	98.1%	0.75	1487	1456	44	13
Matching	0.25	0.516	53.3%	0.25	493	1106	394	1007
	0.50	0.504	51.9%	0.50	448	1108	392	1052
	0.75	0.514	53.6%	0.75	498	1110	390	1002

threshold increases true negatives (TN) at the expense of false negatives (FN).

The ROC plot maps out this balance. For each threshold, it displays the ratio of TP to the sum of TP and FN (True Positive Rate, TPR) along the vertical axis and FP to TN+FP (False Positive Rate, FPR) on the horizontal axis. A classifier with a perfect hit rate will pass exactly through the upper-left corner of the plot at (0,1). A classifier making random decisions will yield a 45-degree line. The Area Under Curve (AUC) summarizes the information in an ROC plot in a single numerical measure between zero and one. AUC corresponds to the probability that the classifier will assign a higher score to a randomly chosen stego image than to a randomly chosen clean image. Thus, an AUC of 1.0 implies that the two classes have no overlap, while 0.5 would mean that the classifier makes random decisions. Since AUC does not rely on a specific threshold, it permits direct comparison between classifiers with scores of different units, like the RS estimate and CNN probabilities used in our experiment.

For the Sequential and Random strategies, detection performance is high across all three embedding rates, with AUC values exceeding 0.98 in every case. At an embedding rate of r = 0.75, both strategies yield near-perfect accuracy (99.8% and 99.9%, respectively). Even at the lowest rate of r = 0.25, the RS analysis achieves AUC values of 0.982 (Sequential) and 0.986 (Random), with accuracies above 95%. The Edge Adaptive strategy proves considerably more difficult to detect. At r = 0.25, the AUC drops to 0.761

and accuracy falls to 72.5%, with 419 false positives and 405 false negatives out of the tested pairs. Performance improves substantially at higher rates, reaching an AUC of 0.997 and accuracy of 98.1% at r = 0.75. The Matching LSB strategy is effectively undetectable by RS analysis. AUC values remain near 0.5 across all embedding rates (0.516, 0.504, and 0.514 for r = 0.25, 0.50, and 0.75, respectively), indicating performance no better than random guessing. The mean RS estimates for stego images under this strategy (0.038 to 0.044) remain close to the clean image baseline of 0.0211, explaining the classifier's inability to distinguish between the two classes.

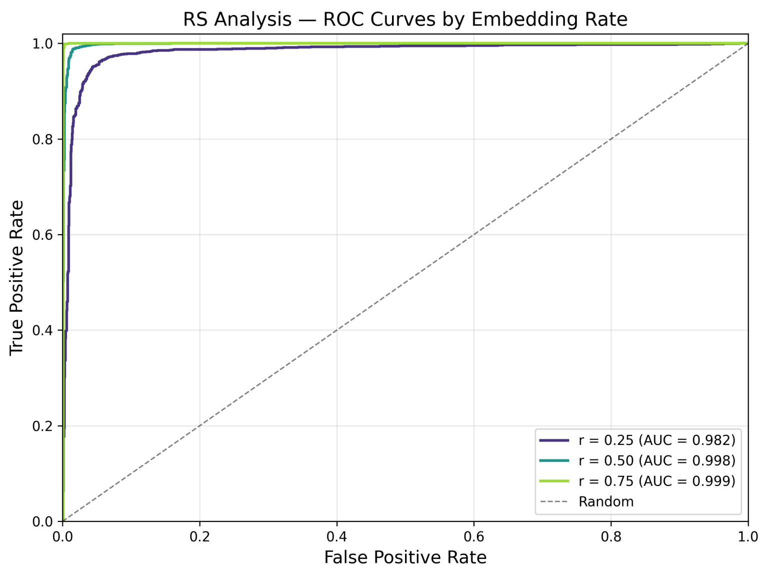

Figure 4.1: RS Analysis: ROC Curves for Sequential LSB embedding at various rates.

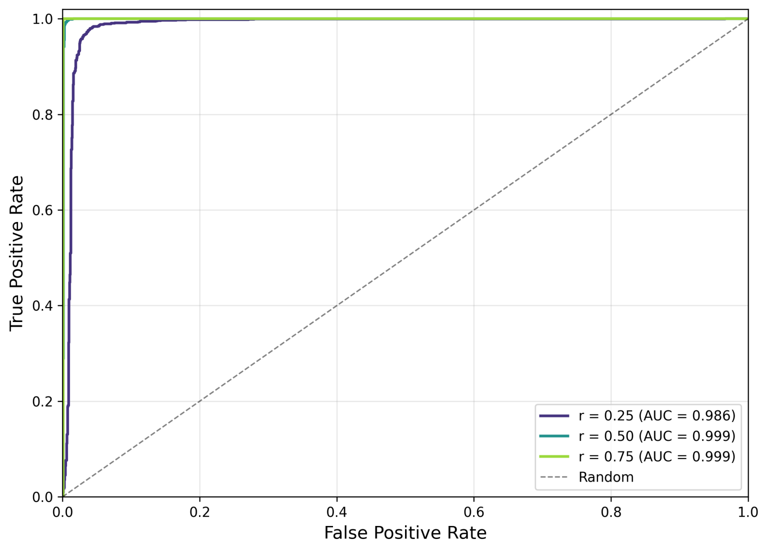

Figure 4.2: RS Analysis: ROC Curves for Randomized LSB embedding at various rates.

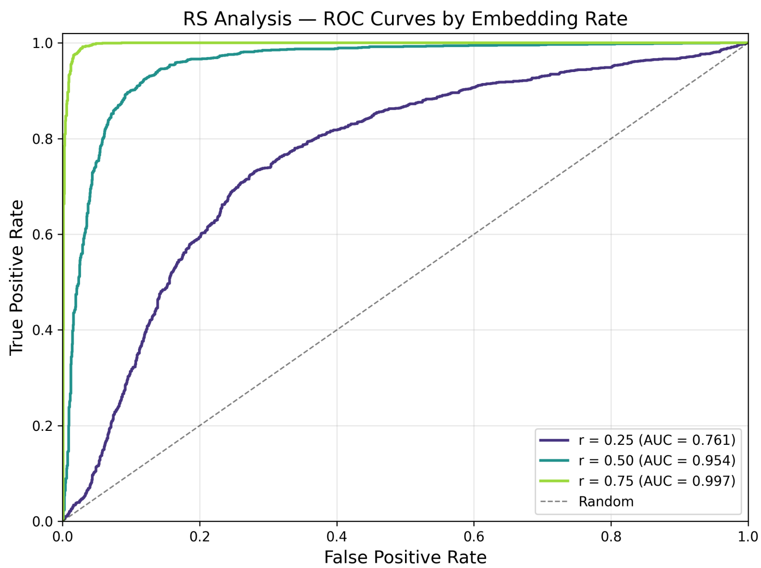

Figure 4.3: RS Analysis: ROC Curves for Edge Adaptive LSB embedding at various rates.

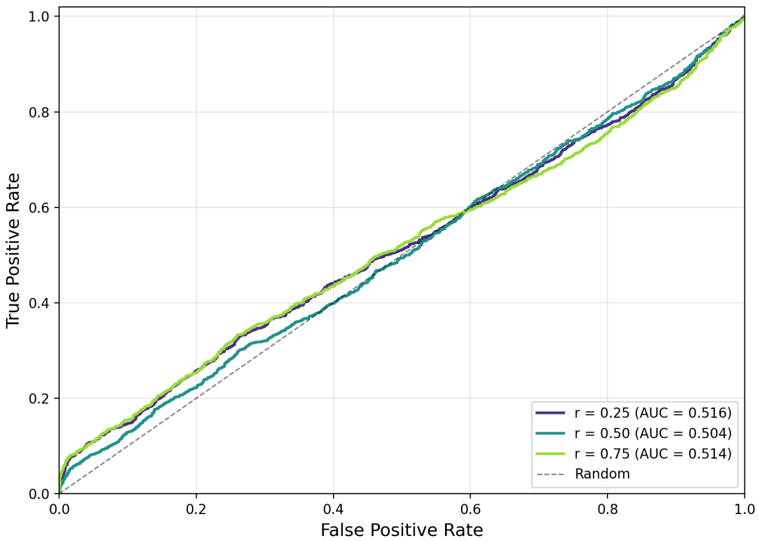

Figure 4.4: RS Analysis: ROC Curves for Matching LSB embedding at various rates.

The corresponding ROC curves are shown in Figures 4.1–4.4. Figures 4.1 and 4.2 illustrate the strong separability achieved for Sequential and Random embedding, with curves closely hugging the upper-left corner. Figure 4.3 shows the progressive improvement for the Edge Adaptive strategy as the embedding rate increases. Figure 4.4 confirms that the Matching LSB strategy produces ROC curves indistinguishable from the diagonal, consistent with random guessing.

Table 4.2: CNN classification performance for different LSB embedding strategies and rates, evaluated on a balanced held-out test set of 3000 images (1500 cover / 1500 stego) per configuration.

Strategy	Rate	AUC	Accuracy	TP	TN	FP	FN
Sequential LSB	0.25	0.960	89.47%	1340	1344	156	160
	0.50	0.994	96.83%	1453	1452	48	47
	0.75	0.999	99.20%	1488	1488	12	12
Random LSB	0.25	0.905	82.37%	1237	1234	266	263
	0.50	0.978	93.17%	1402	1393	107	98
	0.75	0.996	97.63%	1468	1461	39	32
Adaptive LSB	0.25	0.876	79.13%	1195	1179	321	305
	0.50	0.952	88.40%	1335	1317	183	165
	0.75	0.984	94.73%	1427	1415	85	73
Matching LSB	0.25	0.752	68.60%	1042	1016	484	458
	0.50	0.898	82.07%	1245	1217	283	255
	0.75	0.970	91.37%	1380	1361	139	120

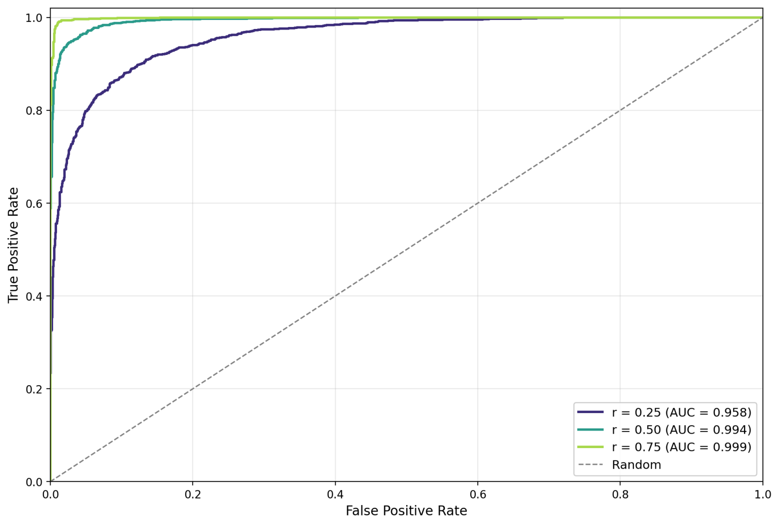

Figure 4.5: CNN: ROC Curves for Sequential LSB embedding at various rates.

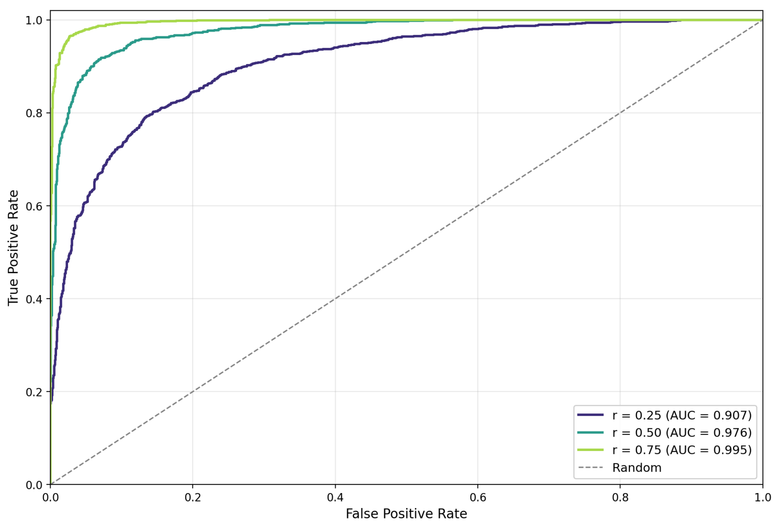

Figure 4.6: CNN: ROC Curves for Randomized LSB embedding at various rates.

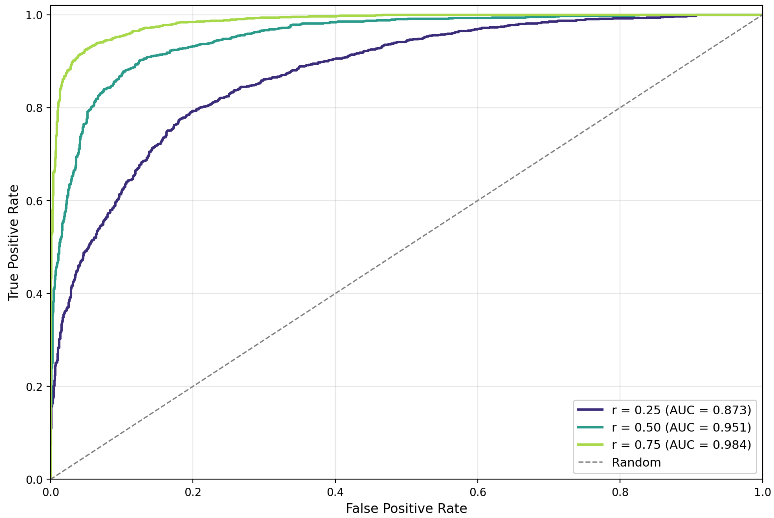

Figure 4.7: CNN: ROC Curves for Edge Adaptive LSB embedding at various rates.

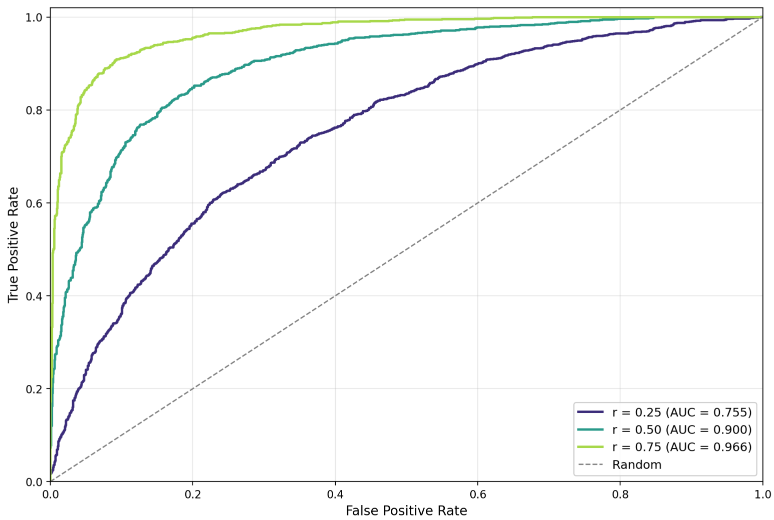

Figure 4.8: CNN: ROC Curves for Matching LSB embedding at various rates.

Accuracy for detection for the trained CNN algorithm is shown in Table 4.2. It is evident from the table that there is a definite hierarchy of detectability: the LSB Sequential method is detected easily, while the Random LSB and Edge Adaptive LSB methods are detected moderately well, with the Matching LSB being the most difficult to detect.

In the case of LSB Sequential Embedding, the CNN obtains 89.47% accuracy at the smallest embedding rate $r = 0.25$, with an increase up to 96.83% at $r = 0.50$, and 99.21% at $r = 0.75$. This high performance should not come as a surprise, since sequential embedding focuses on changes within one block of pixels, resulting in spatially correlated errors.

Detecting the LSB random embedding is a little harder, with an average drop in the accuracy of about 7 percentage points compared to the case of the sequential algorithm for $r = 0.25$, where the accuracy falls down to 82.35%. For larger values of the distortion parameter, this difference becomes smaller, as we have 93.18% for $r = 0.50$ and 97.64% for $r = 0.75$.

Edge Adaptive embedding technique makes detection even more difficult. With $r = 0.25$, the precision drops to 79.12%, which is a decrease by more than 13 percentage points from LSB sequential with the same ratio. The precision increases to 88.41% and 94.73% for $r = 0.50$ and $r = 0.75$, correspondingly. With Edge adaptive embedding, since most alterations are applied to texture-rich areas, where natural variations take place, it becomes more difficult to tell the alterations from the original variation in the picture.

The problem of Matching LSB presents itself as the hardest for the CNN to solve. The CNN achieves an accuracy rate of 68.59% when $r = 0.25$. This accuracy rate is quite high, considering that it is significantly better than the chance accuracy rate of 50%. The highest accuracy rates attained when $r = 0.50$ and $r = 0.75$ were 82.07% and 91.36% respectively. LSB Matching presents its greatest challenge to the CNN because of the basic premise that distinguishes it from other schemes; LSB replacement to LSB Matching adjustment of pixels.

For all four methods, there is an observable trend: accuracy improves linearly with embedding rate, with the biggest gains observed from $r = 0.25$ to $r = 0.50$ (a gain of 7-14%), whereas the gain between $r = 0.50$ to $r = 0.75$ is smaller (3-9%). This diminishing effect can be attributed to the logarithmic dependence of the amount of data on the ability to recognize it statistically;

when doubling the amount of data starting from a very low value provides much more evidence than when doubling from a large amount of data.

5

Discussion

5.1 Interpretation of Results

The main research question asked how a Convolutional Neural Network compares to RS analysis in accurately and reliably detecting LSB steganography in PNG images. Taken as a whole, the results do not support a simple narrative "deep learning is better". They both appear to have different strengths and the preferable method depends on which embedding strategy and payload need to be detected.

For Sequential and Random LSB embedding, the RS analysis performed better at every tested rate. At $r = 0.25$ RS showed 95.4% accuracy on Sequential and 97.0% on Random embedding, while the CNN scored 89.47% on Sequential and 82.35% on Random. The biggest difference between the two steganalysis methods can be seen at low payload embedding (around 15 percentage points in favor of RS), and it decreases at higher rates. These results are aligned with the theory regarding RS analysis, which states that Sequential and Random LSB replacement violate the balance between the number of pixels in R_M and R_{-M} groups, as stated by Fridrich, Goljan, and Du [10] for natural images. This shows that RS analysis is a very good detector of the characteristic feature of this kind of image manipulation, and that generic CNN's have no meaningful advantage over it in this case.

The case for Edge Adaptive LSB is very different. Both methods appear to be performing well at $r = 0.75 (RS = 98.1\%, CNN = 94.73\%)$, but at $r = 0.25$ RS accuracy falls to $72.5\% (AUC\, 0.761)$ while the CNN retains 79.12%.

The adaptive techniques restriction of modifications to textured, noisy regions is the assumption violation that weakens RS analysis. The discrimination function already registers high values in those regions, so LSB flips no longer reliably push groups from Regular to Singular or vice versa. The CNN learns directly from data rather than from an explicit smoothness assumption, and appears to be less affected by this.

The crossover between the two methods on Edge Adaptive LSB embedding thus illustrates the theoretical argument made by Qian et al. [19] and by Apau et al. [1], namely that the rigidity of handcrafted statistical features becomes a liability as embedding techniques begin to respect natural image statistics.

The largest difference can be observed by looking at the tests on LSB Matching. The RS analysis is very close to flipping a coin, with AUCs of 0.516, 0.504 and 0.514 it is virtually a random classifier. This is not a surprising outcome, as Ker [14] observed the ± 1 adjustment in LSB Matching does not flip bits deterministically but rather disturbs pixel values symmetrically. This does not break the $R_M \approx R_{-M}$ assumption that RS analysis relies on. The CNN accuracy ranged from 68.59% at $r = 0.25$ to 91.36% at $r = 0.75, 68.59\%$ accuracy is not great, however, it is a good bit above a random coin flip and demonstrates that the network can learn on artifacts that RS analysis is not usable for. For the Matching LSB method the comparison demonstrates which method is applicable at all rather than which method performs better.

5.2 Sub-Question: Sensitivity to Payload Size

Our sub-question asked how detection performance varies as the hidden payload decreases. Degradation in performance can be observed across both methods, but the shape of the degradation differs enough to be informative.

The degradation of RS analysis appears to be dependent on the embedding technique used to hide the payload. It appears to be performing well for Sequential and Random embedding, as the drop from $r = 0.75$ to $r = 0.25$ is only about four to five percentage points of accuracy. Thus, RS analysis appears to be highly reliant at the lowest tested embedding rates.

In the case of Adaptive embedding, the same decrease in payload results in an even larger drop in performance (from 98.1% to 72.5%), while for Matching embedding there is virtually no impact from varying the payload parameter because the detector does not detect anything at all. This corresponds to what we found out in 2.1.4 where the performance of statistical tests decreases due to the fact that the introduced asymmetry is less than the noise level.

The CNN is also consistent across degradation rates, with an average decrease of 7-14% from $r = 0.50$ to $r = 0.25$ and a smaller decrease from $r = 0.75$ to $r = 0.50$, but there is no catastrophic failure in any embedding scheme. Even on Matching LSB, where RS fails completely at $r = 0.25$, the CNN maintains some discriminative capacity. This suggests that the claim by Apau et al. [1], that learned representations generalize better to low payload embedding is correct, even if in this study, it is only demonstrated for the two more advanced schemes.

Overall, the two approaches provide different answers to our sub-question, depending on the embedding scheme. For Sequential and Random LSB, both approaches can be trusted to work down to $r = 0.25$, but RS is slightly more accurate. On Adaptive LSB, the CNN exhibits less catastrophic degradation, making it the preferred detector below $r = 0.50$. Finally, for Matching LSB, only the CNN shows any degradation at all, since RS cannot distinguish between signals at any rate.

5.3 Relation to Previous Work

Three common ideas in existing research on steganalysis are reaffirmed in these results. As previously established by Fridrich, Goljan, and Du [10] and Fridrich and Goljan [9]), RS analysis is an effective means of identifying classical LSB replacement, with this identification remaining effective even when significant rates of embedded data exist. As also predicted by Ker [14], RS-type detectors do not work well with LSB Matching due to an inherent lack of structural compatibility. As suggested by Qian et al. [19], Wang, Pan, and Fan [23] and Apau et al. [1], CNN-based steganalysis methods best generalize across embedding techniques, whereas handcrafted statistical tests cannot.

The study shows that the traditional steganalysis approach (e.g., RS) is more successful on the tested payload rates. It was established that RS is the best option and demonstrates strength in its operating range as steganalysis techniques have broadened due to neural approaches. Thus, inexperienced practitioners may benefit from the use of RS, while a CNN would be useful if an adaptive method or Matching scheme were used to embed information.

Because of the results of this study, one may practice RS to detect plain LSB replacement steganography on a fitted lossless PNG, while a CNN method may be useful.

5.4 Limitations

Numerous limitations apply to the conclusions drawn by this study regarding cover source and image modalities from one cover source (BOSSBase 1.01) and one specific image modality (grayscale information and subsequently re-encoded as PNG). Sedighi, Fridrich, and Cogranne [20] found that the statistical characteristics of the cover image types significantly influenced both embedding and detection performance, therefore, the absolute accuracy obtained from this research should not be extended beyond BOSSBase to other cover sources such as consumer photographs or screen captures without further testing to verify accuracy. In addition, the payload rates that were examined during this research are relatively high compared to other contemporary steganalysis studies (which often go down to 0.1 bpp or lower). Therefore, it will be interesting to see how degradation in performance occurs for the two methods when using even smaller payload sizes for (i.e., < 0.05) and whether the relative advantage of CNN's becomes apparent when the small payload sizes are used relative to other cover sources.

5.5 Future Work

The limitations mentioned above lead naturally to three avenues for future research directions. The first avenue is to expand the range of payloads down to the sub-0.1 bpp range so that researchers are able to evaluate whether or not the degradation of CNN(s) is more graceful at lower payloads. The second avenue is to broaden the set of cover sources, such as JPEG decompressed to PNG, phone images, and synthetic images, to address the cover source dependence highlighted by Sedighi, Fridrich, and Cogranne [20] and to provide a more stringent assessment of the generalization hypothesis in this domain. The third avenue is the comparison of classical LSB-based approaches to content adaptive approaches, including S-UNIWARD and MiPOD, to allow alignment of the comparisons with current steganalysis benchmarking procedures, and to ultimately provide researchers a means of determining if RS analysis has remaining utility in these modern contexts or whether only neural network detectors can be deployed.

In addition to advancing the state of knowledge about CNNs and cover source properties, researchers will also have the opportunity to produce confusion matrices, ROC curves, and error rates by class that are matched to CNN outputs, allowing researchers to conduct proper FPR comparisons at equivalent operating points.

5.6 Broader Impact

The major implication of these findings is that the choice of method in applied steganalysis should depend on the specific threat model and not simply upon an arbitrary choice based on the latest "in vogue" method available. RS analysis remains a very strong, interpretable, and computationally efficient detector for plain LSB replacement embeds, and its above-average performance against high payloads in this analysis continues to endorse its use within forensic pipelines that target such type of content. When the target is an embed created using an adaptive method or Matching LSB, then a machine learning approach must be adopted; the findings of the CNN in the current analysis demonstrate that CNNs can recover usable signals in contexts where RS analysis would be blind, despite falling below the upper bounds of detection thresholds identified within the published literature using more sophisticated architectures and training strategies. Therefore, these two methods should not be viewed as competitors, but as complementary, and a viable detection strategy would be to run both methods in parallel and combine the results from both systems.

5.7 Ethical Considerations

The empirical findings allow for a reflection with more nuance. The results confirm that the simple and computationally inexpensive RS analysis remains sufficient against LSB techniques such as, Sequential LSB, Random LSB. It means that practitioners do not need access to specialized, and expensive, hardware or large training datasets to address a significant share of real-world cases. The result also show potential exploits by adversaries that could use Matching LSB, with which both RS analysis and our CNN struggled with at low payloads ($r = 0.25$). We consider this disclosure acceptable because the embedding strategies examined are long-established in the public literature [21] [14] [16] and because the study contributes no new embedding method. This research makes comparisons using a common dataset (BOSSBase 1.01) used in academia and adheres to the principles of integrity described in Section 3.1.

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